

Chapter 3

HEAT TRANSFER COEFFICIENT

The calculation of the rate of transport of heat in a solid body or medium is based on a well-defined simple law-the Fourier's law. While the law is equally valid for transport of heat in a fluid medium, there are very few physical situations in which heat transfer occurs in a fluid purely by conduction. For the Fourier's law to be applicable, the medium must be stagnant. But a fluid, in which flow of heat is occurring, rarely remains stagnant in practical situations. As a result, heat transfer in a fluid mostly occurs by convection. The use of the 'heat transfer coefficient' greatly helps us in calculating the rates of heat transfer by convection.

3.1. CONVECTIVE HEAT TRANSFER AND THE CONCEPT OF HEAT TRANSFER COEFFICIENT

Convection refers to transport of heat (or mass) in a fluid medium because of the motion of the fluid. Convective heat transfer is, therefore, associated with bulk motion in the fluid medium. *Bulk motion* or *macroscopic motion* or *bulk flow* can be easily distinguished from molecular motion or microscopic motion.

A pragmatic approach to overcome the difficulty is to define a *heat transfer coefficient*, h , such that

$$Q = h A(T_s - T_o) \quad (3.1)$$

where Q is the rate of heat transfer (W), A is the area of heat transfer (m^2), T_s is the surface temperature, and T_o is the bulk fluid temperature. Written in words, Eq. (3.1) may be expressed as

$$\text{Rate of heat transfer} = (\text{Heat transfer coefficient}) \times (\text{Area}) \times (\text{Temperature driving force}) \quad (3.2)$$

Heat transfer coefficient is a *phenomenological coefficient*. It is based on the observation that the rate of convective heat transfer is proportional to the area of heat transfer and the temperature driving force. The proportionality constant is called the heat transfer coefficient. That is,

$$Q \propto A \quad \text{and} \quad Q \propto T_s - T_o$$

or

$$Q = hA(T_s - T_o) = hA\Delta T$$

Therefore,

$$h = \frac{Q}{A(T_s - T_o)} = \frac{q}{T_s - T_o} = \frac{q}{\Delta T} \quad (3.3)$$

where q is the heat flux and ΔT the driving force. So, quantitatively, heat transfer coefficient is equal to the heat flux per unit driving force. Equation (3.1) is also known as the 'Newton's law of cooling'.

It should be remembered that the heat transfer coefficient combines in it the effects of velocity or motion in the medium and also the properties like thermal conductivity, viscosity, density of the medium, etc. Motion or velocity in a medium can be created by application of an external force. For example, an agitator creates motion in the liquid in a tank, or a pump creates the pressure necessary for flow of a fluid through a pipe. Heat transfer in such systems (for example, in an

agitated vessel or in a pipe flow) is said to occur by *forced convection*. However, if the temperature of a particular layer of a fluid is higher than that of an upper layer, a bouyancy-induced motion is generated. Heat transfer augmented by this kind of motion is called *free convection heat transfer*. Forced convection heat transfer coefficients are generally much higher than their free convection counterparts.

In order to determine the heat transfer coefficient h , a large number of correlations have been developed over the years by heat transfer researchers which are applicable to a variety of physical situations and geometries. The more important correlations will be discussed in the next chapter. Here we will consider a few physically realistic and practically important cases and problems involving convective heat transfer. Table 3.1 gives typical values of heat transfer coefficients (for both free and forced convection) for a number of systems.

Table 3.1 Values of heat transfer coefficients under typical conditions

Description	Heat transfer coefficients		
	W/m ² °C	kcal/h m ² °C	Btu/h ft ² °F
1. Air flowing at a velocity of 12 m/s over a flat plate, 2 m in length, at 1 atm, 25°C (Laminar boundary layer flow of air)	23	19.5	4
2. Air flowing at a velocity of 10 m/s through a 1 inch (25 mm) pipe at 1 atm, 200°C (Turbulent pipe flow of air)	65	53.7	11
3. Water flowing at a velocity of 1.0 m/s through a 1-1/2 inch (38 mm) pipe at 25°C (Turbulent pipe flow of water)	3800	3200	650
4. Benzene flowing at a velocity of 1.0 m/s through a 1-1/2 inch (38 mm) pipe at 40°C (Turbulent pipe flow of an organic)	1200	980	200
5. A vertical plate, 1.5 m high, at a temperature of 65°C in contact with air at 25°C (Natural convection on a vertical plate in air)	4.5	4.0	0.8
6. A horizontal cylinder, 13 cm in diameter, at a temperature of 80°C in contact with air at 25°C (Natural convection on a horizontal cylinder in air)	6.0	5.0	1.0

3.2. OVERALL HEAT TRANSFER COEFFICIENT

Heat transfer from one fluid to another separated by an intervening solid wall is of great practical importance. Calculation of the rate of heat transfer in such a case is done by using an *overall heat transfer coefficient*. Two situations are considered below.

3.2.1. Heat Transfer between Fluids Separated by a Plane Wall

We will first consider heat transfer from one fluid to another separated by a plane solid wall of thickness l (Fig. 3.1). The temperatures of the bulk of the fluids on two sides of the wall are T_1 and T_2 ($T_1 > T_2$), and the corresponding heat transfer coefficients between the wall and the fluids are h_1 and h_2 , respectively. The temperatures at the fluid-wall interfaces are T_{w1} and T_{w2} . The thermal conductivity of the material of the wall is k_w .

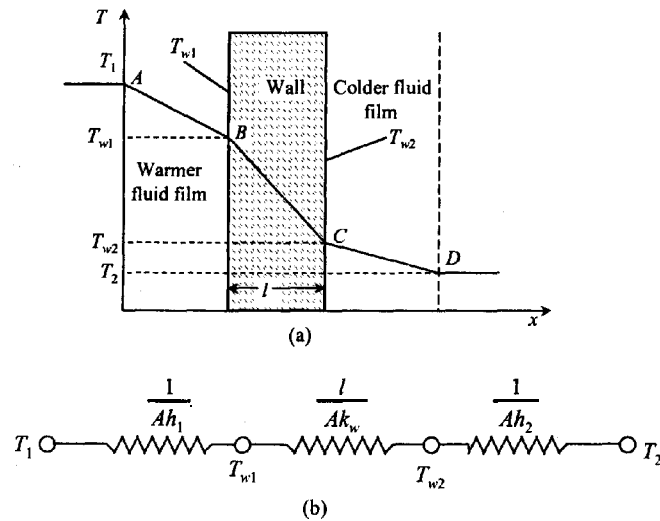


Fig. 3.1 Convective heat transfer between two fluids: (a) temperature profile and (b) electrical analogue

If the area of heat transfer is A , the rates of heat transfer on two sides of the wall are:

$$\text{Rate of heat transfer from the hot fluid to the wall,} \quad Q_1 = h_1 A(T_1 - T_{w1})$$

$$\text{Rate of heat transfer through the wall,} \quad Q_w = k_w A(T_{w1} - T_{w2})/l$$

$$\text{Rate of heat transfer from the wall to the cold fluid,} \quad Q_2 = h_2 A(T_{w2} - T_2)$$

Because heat transfer is taking place at steady state through a constant area, we can write

$$Q_1 = Q_w = Q_2 = Q \text{ (say)}$$

Therefore,

$$T_1 - T_{w1} = \frac{Q}{Ah_1}; \quad T_{w1} - T_{w2} = \frac{Q}{A(k_w/l)}; \quad T_{w2} - T_2 = \frac{Q}{Ah_2} \quad (3.4)$$

Adding the above three relations, we get

$$T_1 - T_2 = \frac{Q}{A} \left(\frac{1}{h_1} + \frac{1}{k_w/l} + \frac{1}{h_2} \right) = \frac{Q}{UA} \quad (3.5)$$

where

$$\frac{1}{U} = \frac{1}{h_1} + \frac{1}{k_w/l} + \frac{1}{h_2}$$

Therefore,

$$Q = UA(T_1 - T_2) \quad (3.6)$$

Comparing Eq. (3.6) with Eq. (2.8), we have

$$\text{Total thermal resistance} = \frac{1}{UA} \quad (3.7)$$

It appears from Eq. (3.6) that the rate of heat transfer has been expressed in a form similar to Eq. (3.1). Here the driving force is the difference of temperature between the hot and the cold fluids, which is, in fact, the *overall driving force* for heat transfer. Here, the quantity U is called the *overall heat transfer coefficient*, and can be calculated if h_1 , h_2 , k_w , and l are known. The overall thermal resistance or heat transfer resistance, $1/UA$, is equal to the sum of the individual resistances—the heat transfer resistances offered by the ‘film’ of the hot fluid ($1/Ah_1$), the wall (l/Ak_w), and the ‘film’ of the cold fluid ($1/Ah_2$).

The temperature distributions in all the three sections (*AB*, *BC* and *CD*) are predictably linear, but have different slopes in general. The electrical analogue of the problem showing the resistances in series is also depicted in Fig. 3.1.

The definition of the overall heat transfer coefficient given by Eq. (3.6) can be used to introduce the concept of *controlling resistance*. If one of the thermal resistance terms is much larger than the sum of the other two, it is called the controlling resistance. For example, if $1/Ah_1 \gg (1/Ak_w + 1/Ah_2)$, then $1/Ah_1$ is the controlling heat transfer resistance. Very often the heat transfer resistance offered by a metal wall ($1/Ak_w$) separating the two fluids is small (because k , is large) compared to $1/Ah_1$ or $1/Ah_2$, and can be, therefore, neglected without causing an appreciable error in the calculation. Also, when heat transfer occurs between a gas phase and a liquid phase separated by a metal wall, the gas phase heat transfer resistance usually becomes significantly larger than the sum of the other two resistances (but may not necessarily be the controlling resistance). This is because the thermal conductivity of a gas is much smaller than that of a liquid.

3.2.2. Heat Transfer between Fluids Separated by a Cylindrical Wall

A well-known simple practical device for heat transfer from one fluid phase to another is the *double pipe heat exchanger*. It consists of two concentric pipes properly fitted or welded with arrangements for pumping one of the fluids through the inner pipe and the other through the annular space. The fluids are thus brought in ‘thermal contact’ in order to achieve heat transfer. A counter-current double-pipe heat exchanger is schematically shown in Fig. 3.2.

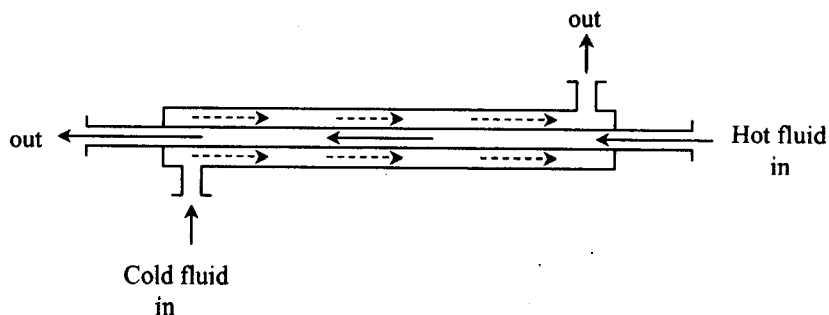


Fig. 3.2 Schematic of a counter-current double-pipe heat exchanger.

How to express the overall heat transfer coefficient for this or a similar device that involves heat transfer in a cylindrical geometry? It can be expected that the requisite equation will be a little different from Eqs. (3.5) and (3.6) because here the wall separating the fluids has a variable area along the radius. Let the inner and outer radii of the inner pipe be r_i and r_o respectively, and the corresponding heat transfer coefficients be h_i and h_o respectively. The inner surface temperature of the pipe wall is T_{wi} and that of the outer surface of the wall is T_{wo} . The bulk fluid temperatures are T_i and T_o ($T_i > T_o$). A cross-section of the device is shown in Fig. 3.3.

We consider a section of the assembly of length L over which the temperatures T_i and T_o are assumed to remain fairly constant. The following rates of heat transfer across the wall of the inner pipe can be written. The rates are equal at steady state.

The rate of heat transfer from the hot fluid to the inner surface (at $r = r_i$) is

$$Q = A_i h_i (T_i - T_{wi})$$

The rate of heat transfer through the pipe wall [see Eq. (2.14)] is

$$Q = \frac{2\pi L k_w (T_{wi} - T_{wo})}{\ln(r_o/r_i)}$$

The rate of heat transfer from the outer surface (at $r = r_o$) of the pipe to the cold fluid is

$$Q = A_o h_o (T_{wo} - T_o)$$

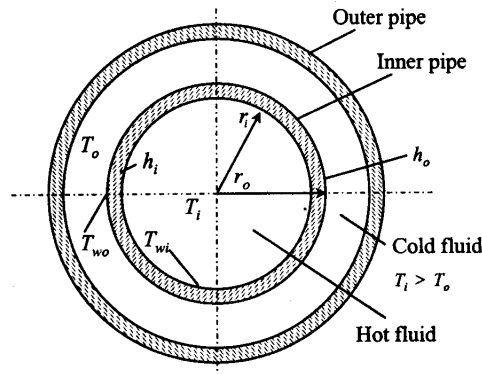


Fig. 3.3 Cross-section of a double-pipe heat exchanger.

Therefore, we have

$$T_i - T_{wi} = \frac{Q}{A_i h_i}, \quad T_{wi} - T_{wo} = \frac{Q}{2\pi L k_w / \ln(r_o/r_i)}, \quad T_{wo} - T_o = \frac{Q}{A_o h_o}$$

where

$A_i = 2\pi r_i L$ = area of heat transfer based on the inner radius of the inner pipe

$A_o = 2\pi r_o L$ = area of heat transfer based on the outer radius of the inner pipe

Adding the above three relations and rearranging, we get

$$Q = \frac{T_i - T_o}{\frac{1}{A_i h_i} + \frac{\ln(r_o/r_i)}{2\pi k_w L} + \frac{1}{A_o h_o}} \quad (3.8)$$

The inside and outside surface areas of the inner pipe are different. So, we should define two overall coefficients - an overall heat transfer coefficient U_i based on the inside surface area and another coefficient U_o based on the outside surface area. However, the rate of heat transfer and the driving force remain the same. Therefore, we have

$$Q = U_i A_i (T_i - T_o) = U_o A_o (T_i - T_o)$$

The above equation, when compared with Eq. (3.8), yields

$$U_i = \frac{1}{\frac{1}{h_i} + \frac{A_i \ln(r_o/r_i)}{2\pi k_w L} + \frac{A_i}{A_o h_o}} = \frac{1}{\frac{1}{h_i} + \frac{r_i \ln(r_o/r_i)}{k_w} + \frac{r_i}{r_o h_o}} \quad (3.9)$$

$$U_o = \frac{1}{\frac{A_o}{A_i h_i} + \frac{A_o \ln(r_o/r_i)}{2\pi k_w L} + \frac{1}{h_o}} = \frac{1}{\frac{r_o}{r_i h_i} + \frac{r_o \ln(r_o/r_i)}{k_w} + \frac{1}{h_o}} \quad (3.10)$$

The denominator of Eq. (3.8) and that of Eq. (3.9) or (3.10) is the sum of the three heat transfer resistances in series. The total thermal resistance R_T is related to U_i and U_o as follows:

$$Q = \frac{T_i - T_o}{R_T} = U_i A_i (T_i - T_o) = U_o A_o (T_i - T_o)$$

or

$$R_T = \frac{1}{U_i A_i} = \frac{1}{U_o A_o} \quad (3.11)$$

and

$$U_i A_i = U_o A_o \quad (3.12)$$

3.3. HEAT TRANSFER FROM EXTENDED SURFACES-THE FINS

Since thermal conductivities of gases are much smaller than those of liquids, the gas-phase heat transfer coefficient is also much smaller than the liquid-phase coefficient. Thus, if we make a heat transfer device for heat exchange between a gas and a liquid, the gas-side ‘film’ will offer most of the thermal resistance, and therefore a large heat transfer area will be required. A practical solution to the problem is to make an arrangement to increase the surface area of heat transfer on the gas side. This can be done by fixing attachments like rectangular metal strips or annular rings, called *fins*, to the surface of heat transfer (e.g. to the surface of a pipe). A cylindrical pipe with a ring-type fin is shown in Fig. 3.4. Because a fin increases the area available for heat transfer, a finned surface is also called an *extended surface*.

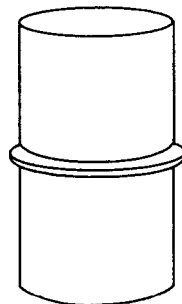


Fig. 3.4 A ring-type fin on a cylinder

However, if a ‘finned’ pipe carries a hot fluid, the temperature at any point of the fin will be less than that of the pipe wall, thus providing a smaller driving force for heat transfer from the surface of the fin. Here we give a theoretical analysis to determine how effective a rectangular fin is in enhancing the rate of heat transfer.

We consider a thin rectangular plate that protrudes a distance l from a wall as shown in Fig. 3.5. This extended surface or fin has the length l , a thickness w and a breadth b . The temperature of the wall is T_w (which is also called the ‘base temperature’ of the fin), and that of the ambient is T_o . Heat loss to the ambient occurs from both surfaces (top and bottom) of the fin. As the thickness of the fin is small, we assume that the temperature at any transverse section of it is uniform. Fin temperature varies only in the longitudinal direction and heat transfer occurs through the fin by conduction in the same direction.

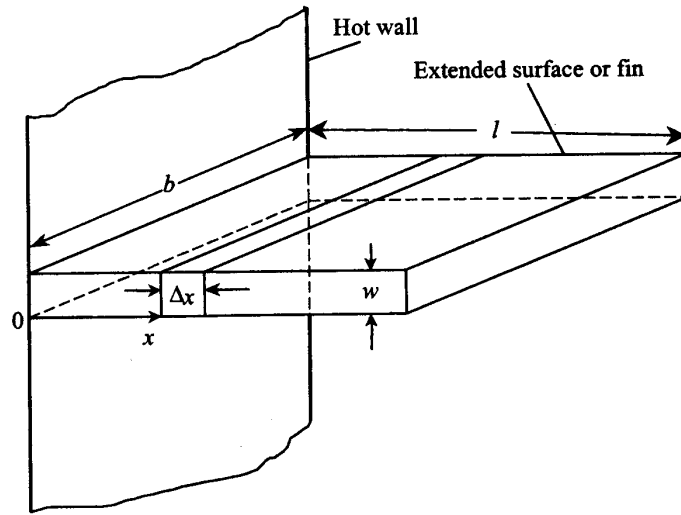


Fig. 3.5 Heat balance in a rectangular fin

Let us now consider a thin section of thickness Δx of the fin at a distance x from the base. For the breadth b of the fin, the available area of heat conduction is bw and the area of heat loss by convection from the exposed fin surface to the ambient is $p\Delta x$ [where p is the fin perimeter = $2(b + w)$]. In order to make a heat balance over the small element, we identify the following heat input and output terms.

$$\begin{aligned} \text{Rate of heat input at } x &= bw q_x|_x \\ \text{Rate of heat output at } x + \Delta x &= bw q_x|_{x + \Delta x} \\ \text{Rate of heat loss from the exposed surfaces} &= p\Delta x h(T - T_o) \\ \text{Rate of heat accumulation (at steady state)} &= 0 \end{aligned}$$

where T is the local temperature of the fin and h is the surface heat transfer coefficient. Making a steady state heat balance, we have

$$bwq_x|_x - bwq_x|_{x + \Delta x} - p\Delta x h(T - T_o) = 0$$

Dividing by $bw \Delta x$ throughout and taking limit $\Delta x \rightarrow 0$, we get

$$-\frac{d}{dx}(q_x) = \frac{ph}{bw}(T - T_o)$$

Putting $q_x = -k dT/dx$, and defining $\bar{T} = T - T_o$, we have

$$\frac{d^2 \bar{T}}{dx^2} = \frac{ph}{kbw} \bar{T} = \xi^2 \bar{T} \quad (\text{where } \xi = \sqrt{\frac{ph}{kbw}}) \quad (3.13)$$

To solve Eq. (3.13) for the temperature distribution in the fin, we need to write down the boundary conditions at $x = 0$ and $x = l$.

$$\text{At } x = 0, \quad T = T_w, \quad \text{i.e.} \quad \bar{T} = (T_w - T_o) \quad (3.14a)$$

$$\text{At } x = l, \quad -k \frac{dT}{dx} = h(T - T_o), \quad \text{i.e.} \quad -k \frac{d\bar{T}}{dx} = h\bar{T} \quad (3.14b)$$

Equation (3.14b) is a *convective boundary condition*. It implies that the rate at which heat is conducted from inside the solid to the boundary must be equal to the rate at which it is transported to the ambient medium by convection.

Solution to Eq. (3.13) is

$$\bar{T} = C_1 \cosh(\xi x) + C_2 \sinh(\xi x) \quad (3.15)$$

where C_1 and C_2 are integration constants.

Using the boundary condition given by Eq. (3.14a), we get

$$C_1 = T_w - T_o \quad (3.16)$$

Differentiating $\bar{T}$ in Eq. (3.15) and using Eq. (3.14b), we get

or

$$-k[C_1 \xi \sinh(\xi l) + C_2 \xi \cosh(\xi l)] = h[C_1 \cosh(\xi l) + C_2 \sinh(\xi l)]$$

$$C_2 = -\frac{C_1 \left[\frac{h}{k\xi} \cosh(\xi l) + \sinh(\xi l) \right]}{\cosh(\xi l) + \sinh(\xi l)} \quad (3.17)$$

Substitution of the relations for C_1 and C_2 in Eq. (3.15) and simplification lead us to the final solution for the *dimensionless fin temperature* given below.

$$\frac{\bar{T}}{T_w - T_o} = \frac{T - T_o}{T_w - T_o} = \frac{\cosh \xi(l - x) + \frac{h}{k\xi} \sinh \xi(l - x)}{\cosh(\xi l) + \frac{h}{k\xi} \sinh(\xi l)} \quad (3.18)$$

It may be seen from the solution [Eq. (3.18)] that the temperature decreases with distance x from the base. So the surface heat flux from the fin also decreases in the longitudinal direction because of a smaller temperature driving force. It is easy to visualize that if the thermal conductivity k of the material of the fin is extremely high, its thermal resistance will be negligibly small and the temperature will remain almost uniform throughout at T_w , which is the wall or base temperature. The maximum rate of heat transfer can be achieved under this condition, which is rather an ideal condition. It is, therefore, interesting and useful to calculate the actual rate of heat transfer and also its ratio to the maximum rate of heat transfer.

The actual rate of heat loss from the fin must be equal to the rate of heat input by conduction at the fin base from the wall at steady state. Therefore, we have

$$Q = (\text{cross-sectional area})(\text{heat flux at the fin base}) = (bw) [-k(dT/dx)]_{x=0}$$

The temperature gradient dT/dx can be determined from Eq. (3.18). Therefore, we get

$$Q = bw(T_w - T_o)(-k) \left[\frac{-\xi \sinh \xi(l - x) - \frac{h}{k\xi} \cdot \xi \cosh \xi(l - x)}{\cosh \xi l + \frac{h}{k\xi} \sinh \xi l} \right]_{x=0} \quad (3.19)$$

If the fin temperature remains uniform at T_w (i.e. the base temperature), the rate of heat loss is the maximum.

$$\begin{aligned} Q_{\max} &= (\text{total area})(\text{heat transfer coefficient})(\text{driving force}) \\ &= (pl + bw) (h) (T_w - T_o) \end{aligned} \quad (3.20)$$

The fin efficiency η is defined as

$$\eta = \frac{\text{Actual rate of heat transfer}}{\text{Calculated rate of heat transfer if the temperature of the fin remains uniform at the base value, } T_w}$$

Therefore, from Eqs. (3.19) and (3.20) we get

$$\eta = \frac{Q}{Q_{\max}} = \frac{bwk\xi}{h(pl + bw)} \frac{\sinh \xi l + \frac{h}{k\xi} \cosh \xi l}{\cosh \xi l + \frac{h}{k\xi} \sinh \xi l} \quad (3.21)$$

A simpler but nevertheless accurate expression for heat loss can be obtained by using an alternative form of the boundary condition Eq. (3.14b). If we assume that heat loss from the edge of the fin (at $x = l$) is negligibly small, we can write

$$x = l, \quad -k \frac{dT}{dx} = 0 \quad (3.22)$$

Solution of Eq. (3.13) subject to boundary conditions (3.14a) and (3.22) can be obtained as

$$\frac{T - T_o}{T_w - T_o} = \frac{\cosh [\xi(l - x)]}{\cosh \xi l} \quad (3.23)$$

which is the longitudinal temperature distribution in the fin. The rate of heat loss from both the surfaces (neglecting that from the edge) is

$$Q = (bw) \left[-k \frac{dT}{dx} \right]_{x=0} = bwk\xi \frac{(T_w - T_o) \sinh \xi l}{\cosh \xi l} \quad (3.24)$$

The fin efficiency is given by

$$\eta = \frac{Q}{Q_{\max}} = \frac{bwk\xi}{plh(T_w - T_o)} \frac{(T_w - T_o) \sinh \xi l}{\cosh \xi l} = \frac{1}{\xi l} \tanh \xi l \quad \left(\because \xi^2 = \frac{ph}{kbw} \right) \quad (3.25)$$

If the fin thickness w is small, $p = 2(b + w) = 2b$, then $\xi^2 = 2h/kw$. It is easy to check the accuracy of Eq. (3.25) for a physically realistic situation. If we take $l = 5$ cm, $w = 4$ mm, $b = 1$ m, $k = 160$ W/m °C, $h = 50$ W/m °C, then $\xi = [ph/kbw]^{1/2} = 12.5$ m⁻¹, and $\xi l = 0.625$. From Eq. (3.21), $\eta = 0.88$; and from Eq. (3.25), $\eta = 0.887$. These values of η are very close to each other. Equation (3.25) is good enough for practical purposes.

Fins may be of different types: longitudinal fins with (a) rectangular, (b) triangular, (c) concave parabolic, (d) convex and parabolic, and (e) radial profile [Fig. 3.6(i)]. Other types include thin annular ring, cylindrical spine, conical spine, etc. External and internal longitudinal fins on pipes are shown in Fig. 3.6(ii). Fin efficiency may be calculated for each geometry of the fin. Choice of the shape and the material of a fin depend upon factors like the material cost, equipment volume, etc.

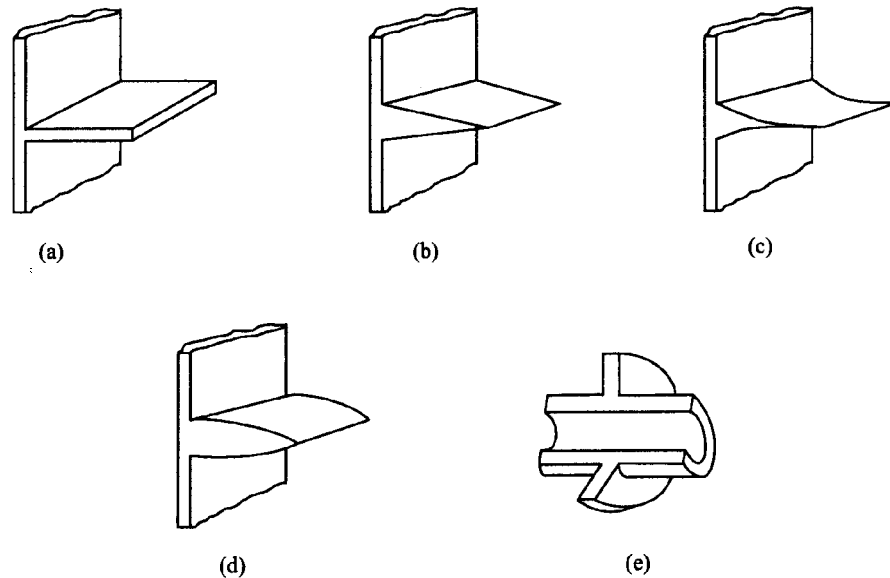


Fig. 3.6(i) A few common types of fins: (a) rectangular, (b) triangular, (c) concave parabolic, (d) convex parabolic, and (e) radial.

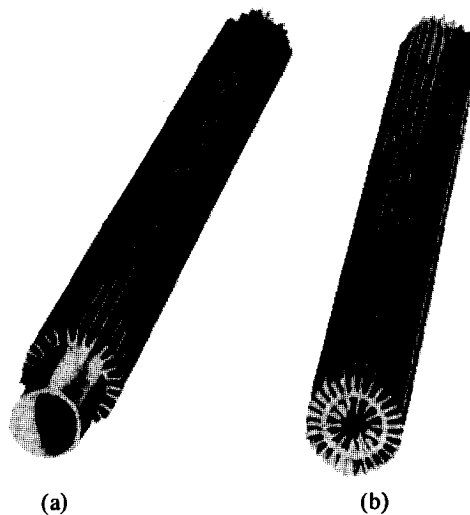


Fig. 3.6(ii) Longitudinal fins on pipes: (a) external and (b) both external and internal

3.4. THERMAL CONTACT RESISTANCE

When two flat solid surfaces are in contact, there may be scattered tiny gaps containing an entrapped gas (usually air) at the contact surface. These gaps, which prevent perfect contact of the surfaces, arise out of surface roughness of the solids (Fig. 3.7). The resulting heat transfer resistance at the interface is called the *thermal contact resistance*. If the surfaces are optically plane, a perfect contact is possible. There would be no gap, making the contact resistance zero.

Thermal contact resistance at a solid-solid boundary is usually expressed in terms of a *contact heat transfer coefficient*.

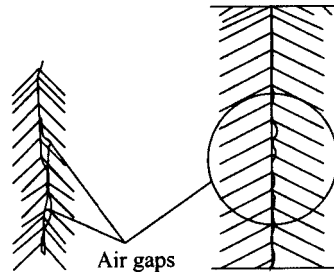


Fig. 3.7 Schematics of surfaces at contact.

3.5. CRITICAL INSULATION THICKNESS

Heat loss from an insulated cylindrical surface is pretty common in industrial operations, for example, heat loss from an insulated steam pipe. Here there are two major resistances in series—the resistance offered by the layer of insulation, and that offered by the ‘gas-film’ on the outside. While the use of a thicker layer of insulation increases the resistance to heat transfer, it also increases the outer radius and hence the area of heat transfer. Thus two factors, having opposite effects on the rate of heat transfer, come into play when the thickness of insulation on a pipe is increased. In some cases this results in a higher rate of heat transfer with increasing thickness of insulation. It is interesting to determine the insulation thickness for which the rate of heat transfer will be a maximum.

We consider a pipe (carrying steam, for example) with a layer of insulation on it as shown in Fig. 3.9. The inner radius of the layer of insulation and the corresponding temperature are r_i and T_i , respectively, and the outer radius of the insulation is r_o . The ambient temperature is T_o , the convective heat transfer coefficient outside the insulation is h_o , and the thermal conductivity of the insulating material is k_c . The heat transfer coefficient, on the inner side of the insulation, h_i , is assumed to be large. The rate of heat transfer from a section of length L of the insulated pipe can be written following Eq. (3.8).

$$Q = \frac{T_i - T_o}{\frac{\ln(r_o/r_i)}{2\pi k_c L} + \frac{1}{A_o h_o}} = \frac{2\pi L(T_i - T_o)}{\frac{\ln(r_o/r_i)}{k_c} + \frac{1}{r_o h_o}} \quad (3.26)$$

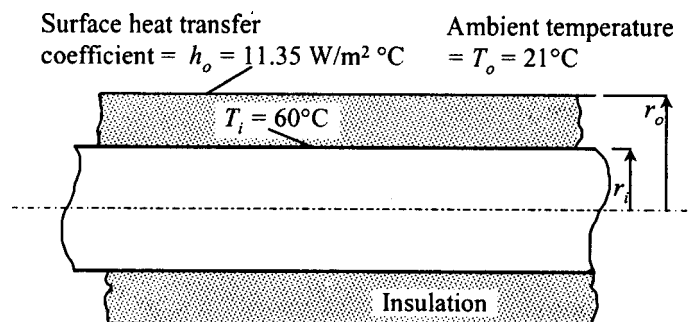


Fig. 3.9 Heat loss from an insulated pipe

We want to determine the outer radius of insulation for which the heat loss will be a maximum. The condition for this is

$$\frac{dQ}{dr_o} = \frac{-2\pi L(T_i - T_o) \left(\frac{1}{k_c r_o} - \frac{1}{h_o r_o^2} \right)}{\frac{\ln(r_o/r_i)}{k_c} + \frac{1}{r_o h_o}} = 0$$

or

$$\frac{1}{k_c r_o} - \frac{1}{h_o r_o^2} = 0$$

or

$$r_{o, \text{crit}} = \frac{k_c}{h_o} \quad (3.27)$$

It can be easily verified that the second derivative, d^2Q/dr_o^2 , is negative and hence Eq. (3.27) gives the condition for the *maximum* rate of heat transfer. The quantity, $r_o - r_i$, is called the ***critical insulation thickness***.

Once r_o is determined from Eq. (3.27), the maximum rate of heat loss can be calculated from Eq. (3.26). When an insulation is applied to a steam pipe, its thickness should be above the critical value. For an insulated electrical wire, however, the insulation thickness should be very near the critical value so that heat loss is maximum, and the insulation temperature remains low thus enhancing the life of the insulation.

It should, however, be noted that a critical insulation thickness may not always exist for an insulated pipe. For example, if the values of k_c and h_o are such that the ratio k_c/h_o ($= r_o$) turns out to be less than r_i (which is the radius of the bare pipe), then the critical insulation thickness, $r_o - r_i$, becomes negative. This means that the critical insulation thickness does *not exist* in this case.

PROBLEMS

- 3.1. An 8% solution of alcohol from a fermenter is raised to 75°C in a heat exchanger before feeding it to the distillation column for concentration. A 50 mm (schedule 40 IPS) pipe is used to carry the hot liquid to the column. The pipe is insulated with a 4 cm thick layer of glass fibre blanket ($k = 0.068 \text{ W/m }^{\circ}\text{C}$). The liquid-side film coefficient is estimated to be $800 \text{ W/m}^2 \text{ }^{\circ}\text{C}$, and the outer air-film coefficient is $10 \text{ W/m}^2 \text{ }^{\circ}\text{C}$. The following data are also available: ambient air temperature = 28°C , actual pipe diameter: i.d. = 53 mm, o.d. = 60 mm. Thermal conductivity of pipe wall is $45 \text{ W/m }^{\circ}\text{C}$, Calculate
- the overall heat transfer coefficient based on the i.d. of the pipe,
 - the overall heat transfer coefficient based on the o.d. of the insulation,
 - the percentage of the total resistance offered by the 'air-film',
 - the rate of heat loss per metre length of the pipe, and
 - the insulation 'skin temperature'.
- 3.2. An aluminum fin ($k = 200 \text{ W/m }^{\circ}\text{C}$) 3.0 mm thick and 7.5 cm long protrudes from a wall. The base is maintained at 300°C , and the ambient temperature is 50°C with $h = 10 \text{ W/m}^2 \text{ }^{\circ}\text{C}$. Calculate the heat loss from the fin per unit depth of material.
- 3.3. Pre-stressed multi-layered shells are often used for high pressure services (the urea reactor is a typical example) because of a smaller wall thickness required compared to a single-layer shell to withstand the same internal pressure. Such shells are often made by 'shrink fitting'. As such there is a possibility of existence of a thermal contact resistance at the interface between the two layers because of the entrapped gases in the crevices on the metal surfaces.
- A two-layered cylindrical shell of high tensile steel (Cr-Mo-V) has an i.d. of 90 cm, and an o.d. of 110 cm. The layers are equal in thickness. Under steady operating conditions, the inside temperature of the shell is 180°C and that of the outer surface is 170°C , making a temperature drop of 10°C across the composite wall. The thermal conductivity of the alloy is $37 \text{ W/m }^{\circ}\text{C}$, and the rate of heat loss through the wall is known to be 5.18 kW per metre length of the shell.
- Is there any thermal contact resistance at the interface between the layers?
 - If so, calculate the contact resistance based on one metre length of the shell and also express it in terms of a contact heat transfer coefficient.
 - Calculate the 'temperature jump' across the contact surface.
- 3.4. Calculate the critical radius of insulation for asbestos ($k = 0.17 \text{ W/m }^{\circ}\text{C}$] surrounding a pipe and exposed to room air at 20°C with $h = 3.0 \text{ W/m}^2 \text{ }^{\circ}\text{C}$. Calculate the heat loss from a 200°C , 5.0-cm-diameter pipe when covered with the critical radius of insulation and without insulation.